

Section 1

New Rules of Business

Chapter 2

Digital Transformation in Business: An Introduction

Key Takeaways

What is digital transformation in business?

Five steps to digital transformation

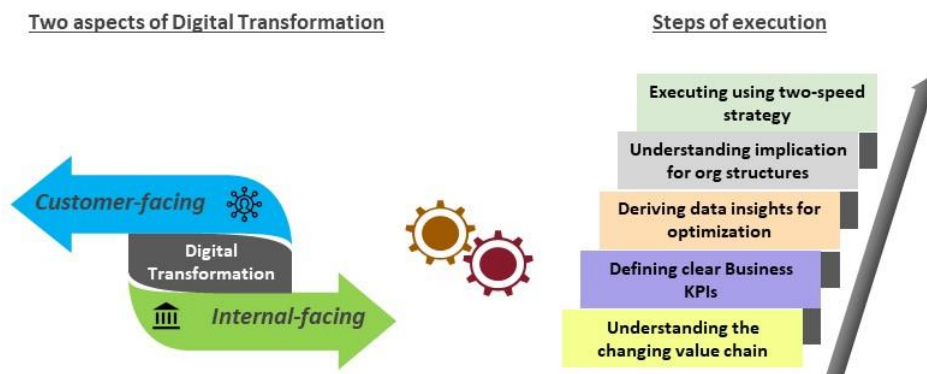
Who should lead digital transformation?

How to lead digital transformation?

The technologies enabling digital transformation

Lesson 1: What is digital transformation in business?

Digital Transformation - An overview



Digital transformation is a very broad topic and it covers a number of different types of initiatives. But what it boils down to is changing how a business interacts with customers using technology, and how its internal processes can be significantly improved by using technology. So, digital transformation in business has two aspects: customer-facing and internal. For customers, it is about providing them a range of technology-based channels for engagement, sales, service and other interaction needs. Internally, it is typically about digitizing or automating processes that can substantially improve both efficiency and response time. This dual nature of digital makes it truly transformative – it can dramatically improve access and experience for customers while reducing response time and costs of delivery for enterprises.

Getting off the starting block

To be successful with digital transformation, the starting point has to be the customer. Digital transformation starts with knowing who your customer is, how their needs are changing and what you need to offer them.

To understand this, let's look at the example of the investment management industry. Historically, an investment management company would sell its investment products through advisers. But now, in the digital economy, there is the opportunity to sell to the investor (the consumer, directly), bypassing the adviser. This is a good example of a shift that digital economy can cause.

Lesson 2: Five steps to digital transformation

Step 1. Understanding how your industry value chain is changing, who your customer is and what you are offering to them. In the example above, the customer is changing from the intermediary – that is, the adviser – to the direct consumer. That's what businesses, in this case, investment management firms, need to think about: how do I directly engage with the consumer, what are his or her needs, and what is it that I can offer him or her?

Step 2. Anchoring the digital transformation in Business KPIs (key performance indicators). Digitization is all-pervasive, and that's a huge opportunity as also a problem. It's very easy to get lost in digital transformation because of the multi-functionality and complexity of these initiatives. It is necessary, therefore, to anchor these initiatives by defining clear business KPIs. Is it about revenue growth? Is it about higher customer engagement? Is it about customer retention? Is it about product up-selling? Having numbers with which you can anchor the digital journey will keep you focused, otherwise, it is easy to lose focus in a digital transformation initiative.

Step 3. Recognizing data as the secret sauce of the digital world. Digital channels generate massive new volumes and a variety of data about customers. This data provides a great opportunity for personalization – for each customer you can optimize offerings and interactions appropriately. I asked one of my clients, one of the largest telecom companies in the US, how they looked at segmentation.

The response was: 'We now have 130 million segments' – because each customer was now well identified.

'You can identify every one of your customers and customize your offerings for each too. That's the great possibility with digital, and that is really made possible through data. That's the third step: clarifying and specifying the customer problem to be solved, identifying what data is needed to solve that problem, and how do you make that data available.'

Step 4. Understanding the implications for organizational structures. By its very nature, digitization tends to be cross-functional; it is not just about IT or marketing, it's about many of these functions coming together. The question is: how do you organize your business so that you're able to execute on these cross-functional projects? And finally, you execute on digital transformation initiatives with:

Step 5. The two-speed strategy: If you implement digital projects with a big bang effort, chances of failure are high. While the promise of digital is very high, the success rate is not very high. The way to execute digital projects is what I call *the two-speed strategy*, which is to stay focused on specific use cases within an organization (small wins). Don't make it all-pervasive. Focus on simple but targeted use cases, leverage these quick wins as a pilot or foundational platform and build on them, as opposed to trying to do it all or launching a project across the whole organization.

Summary:

To begin a digital transformation project, this is how you can start:

- Identify customer problems;
- Define business KPIs;
- Figure out what data is required;
- Bring different functions across the organization together;
- Start with very specific use cases.

Lesson 3: Who should lead digital transformation?

Before addressing how digital transformation must be led, it is important to understand who should lead digital transformation in the enterprise, because this is becoming an awkward problem in many organizations.

Should the CEO, CMO or CIO/CTO lead the change, or should a new role be created? This is a tough choice, one that often creates friction. The answer is, no one person can really lead digital transformation fully, because digitization cuts across marketing, technology, operations and finance. As I said earlier, it is pervasive, and one person trying to lead an enterprise's digital transformation is a recipe for disaster.

Creation of new roles doesn't wholly address the challenge either. For example, the role of chief digital officer is emerging, but it is often difficult to fully empower this role. The CIO and the CMO roles still exist, so the chief digital officer would still depend on them, and the role ends up being an almost superficial one. Those approaching the issue of digital transformation leadership should look towards the digital natives – like Google, Facebook, and Amazon – and learn how they have done it.

These digital natives would never say that they are going through a digital transformation journey and that their CIO or CMO is leading it. The entire company is organized around delivering on customer problems. The product manager would be the one leading specific initiatives for different customer problems. In Google, you will see hundreds of product initiatives and hundreds of product managers. Its organization is structured around customer problems. I strongly believe that's the way to go; that's the way legacy enterprises can really succeed in the digital world. The heart of their problem with digital lies in their organizational structure. Enterprises need to reorganize themselves into this product-based structure, which is a cross-functional structure and is organized around customer problems. This is how the digital natives have done it, and I think the time has come for legacy enterprises to adopt the same philosophy.

Lesson 4: How to lead digital transformation

There are a number of key attributes that leaders (or product managers) need to possess to make a success out of digital transformation. Here are two:

The first is **deep customer insight**. In any business, the customer is king. In the digital economy, this philosophy becomes even more critical. A deep understanding of the customer – what customers need now and what they will need in the future – is the most important starting point to lead digital transformation. The second is the ability **to understand both technology and business**. This will require a new breed of leaders or managers who are equally comfortable with technology and business. This tends to be an issue in large legacy enterprises where the IT function doesn't necessarily understand the business well, and where there are business functions that don't understand IT.

Managerial ability to bridge these two worlds is essential, and it is becoming a prerequisite for roles such as the CIO, CTO and product manager. There are more aspects to this, but deep customer understanding and the ability to bridge both IT and business must be the key attributes of product managers leading digital transformation.

Lesson 5: The technologies enabling digital transformation

There are four key technologies enabling digital transformation: data science, AI and machine learning, automation, and cloud. These are the four pillars of digital, and each has a certain role to play in an enterprise's digital transformation.

Data science is at the heart of digitization. Data science is what makes possible optimization of specific customer journeys and internal processes.

The power of AI and machine learning is at the core of personalization, necessary for enhancing customer experience at the individual level. With machine learning, the role of data is moving from one of insight to action.

Automation brings a lot more efficiency to internal processes. As mentioned earlier, digital transformation for the business as a whole has implications for internal as well as external processes. Optimization of internal processes is where automation technologies – like RPA and intelligent automation – are making a big difference.

Finally, the **cloud enables digital transformation**. Initially cloud's role was more on the infrastructure side. Embracing cloud makes the entire organizational infrastructure and all its applications a lot more scalable. With digital, your demand could grow very significantly, and cloud provides you a scalable and efficient way of building your infrastructure. Cloud's role is now expanding beyond infrastructure and it is becoming a key enabler of transformation across the technology stack.

Nitin's Conclusion

***Digital transformation should be a strategic priority.** In the digital economy, customers are getting a very different value proposition from what they did before. If businesses do not shape up, it becomes a question of their own survival and possible extinction, especially given the current situation, with the coronavirus pandemic having forced businesses to embrace digital alternatives to their usual physical operations. We have seen it very vividly play out in the retail industry when the likes of Kmart and even Walmart in the US and all over the world were severely challenged by Amazon. That's a trailer of the movie that's likely to play out in industry after industry, simply because the customer is getting a value proposition from these digital natives which is of an order of magnitude superior, lower cost, and of higher convenience and better customer service than from their old-world counterparts. If, as a legacy enterprise, you do not embrace digital transformation, both externally for your customers and internally for optimizing your processes, you will be dead.*

One of the key ways to ensure success in digital transformation for the legacy enterprise is to connect strategy and execution. The way service companies providing digital transformation services are structured today, they have either strategy consulting capabilities or specialize as IT execution shops. This is the problem. Digital is a space where, if strategy and execution are not integrated or connected, chances of execution failure are very high!

Section 1

New Rules of Business

Chapter 3

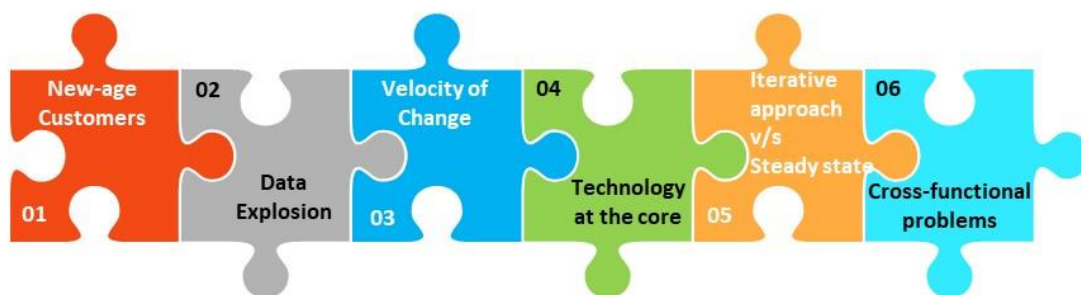
Framework for Digital Transformation, Part One - Six Reasons why Digital Demands a New Business Paradigm

Key Takeaways

Why companies need a new business paradigm

Lesson 1: Six Reasons why Digital Demands a New Business Paradigm

Six Reasons why we need a new business paradigm for the Digital Age



New-age customers:

Today's digital customers are younger, more fickle, and typically in the eighteen-to-thirty-five 'millennial' demographic. They are not only active on social media but also greatly influenced by their peers and the brands they follow across various platforms like Twitter, Facebook, Instagram and Snapchat. Eighty-six per cent of millennials say they use social media, compared with smaller shares among older generations.⁶ Because of this, they are not as brand loyal as the older generations and are extremely price sensitive. At the same time, they are more demanding and expect high-quality digital experiences.

An explosion of data:

The volume, velocity and variety of data that we are seeing today are unprecedented and will continue to grow exponentially, especially for digital businesses. Research by Datareportal has shown that more than half of the world's population now uses social media, with more people using social media than not.⁷ This enormous amount of data gives businesses a phenomenal opportunity to understand customers deeply and offer them more personalized experiences. This is becoming the norm and is now generally expected by most of the customers, prospects and peers we interact with today. On the flip side, data can also become quicksand – in the sense that if an organization doesn't harness it for the customer and organizational benefit efficiently, it can suck the firm into a never-ending investment cycle.

Increased velocity of change:

The velocity of change has increased on all fronts. Technologies from a few years ago, like cloud and mobile, have become commonplace and have been replaced by newer technologies like AI, the internet of things and blockchain, which are driving current conversations. In a couple of years, these technologies will also be replaced by something else. The velocity of change is equally rapid on the business front. Industry structures are shifting, and new competitors are emerging as a result. Here's a hypothetical example: Let's say you run a large global bank and you're introducing a digital strategy. You're not only concerned about what your competitor bank is doing, but you're also now waiting to see how a disruptor like Amazon is breaking into the financial services world. Another reality is that your traditional markets probably don't offer you the same opportunities they did earlier; new markets with much higher growth rates are constantly emerging.

Technology at the core:

Technology started off as a support function that ran company servers and data centres. Over the past decade or so, it has progressed to becoming an enabler that builds and maintains the applications that run businesses. Today, technology is front and centre – in the sense that a digital business is a technology business. To succeed in this new digital reality, every company needs to reinvent itself and build a technology DNA. If organizational leaders do not truly understand the power of technology and what it can do for their company, their digital business will not succeed. Equally, the capabilities and profile of the technology function need to be significantly upgraded as we now need the technology functions to be transformational leaders.

Iterative versus steady-state approach:

There are three themes working in conjunction to make digital transformation a continuously moving target requiring an iterative approach. First, customers have ever-evolving needs. Second, there is an explosion of data and constant improvement in analytics capabilities that allow you to continually deliver superior insights. Third, rapidly maturing digital technologies enable continuous deployment, taking away the need for big-bang implementations. As a result, 'right the first time' does not exist; your whole approach has to become iterative. Continuous experimentation and learning are key. You execute, get new insights, tweak, improve, then re-execute.

Cross-functional problems:

When it comes to building a digital business, the problem is not simply about marketing, finance, operations or technology; digital problems are interdisciplinary. Thus, it is about how all of these different functions come together in a cross-functional set-up to develop a solution to customer problems. In my experience, this requires greater organizational collaboration and multiple functions that work seamlessly. Digital natives tend to be more successful in accomplishing this collaboration, while legacy enterprises tend to struggle to go beyond their embedded organizational structures.

Nitin's Conclusion:

The bottom line is that the fundamental nature of a digital business is very different from that of a traditional business. Until big enterprises understand this, I believe they will struggle to get digital right. Expecting to build a successful digital business without changing the organizational fundamentals is a futile pursuit. It is not about building a digital business but about building an end-to-end business transformation.

Section 1

New Rules of Business

Chapter 4

Framework for Digital Transformation, Part Two - Five Rules for Successful Digital Transformation

Key Takeaways

Five action points to achieve transformative success

Lesson 1: Five Rules for Successful Digital Transformation

Five rules for successful Digital Transformation

 **Digital Fundas**
A CSR initiative of Incedo



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Rule 1: Rethink the business model

Industry value chains are becoming disintermediated. The end customer, who was once a few steps removed, now interacts with your enterprise directly. You need to rethink your business model to benefit from this disintermediation and to engage with this younger, digitally-savvy customer. You also need to rethink your products, pricing, customer touch points, service delivery models, etc. As you evolve the business models, you cannot be afraid of cannibalizing some of your existing activities or business. Business model decisions are often limited by the worry that some of your revenue from an existing channel or service line might be impacted. This incremental thinking does not fit in the digital age. If you are not ready to cannibalize a portion of your business for your own long-term and larger benefit, someone else will come past and cannibalize you.

Rule 2: Focus obsessively on business and customer outcomes

Digital transformation initiatives can end up becoming massive, never-ending projects, and it is easy to lose sight of what one set out to achieve. In my experience, the ideal way to start is with sharply defined key performance indicators (KPIs) that will have the most material impact on your business. Once the business KPIs are defined, they should become the anchor for the entire transformation programme. Business KPIs should flow from your strategic priorities, such as increasing the digital channel mix, increasing cross-selling or up-selling or improving customer retention. Once you have chosen the business KPIs as a starting point, the next step is to identify and solve the customer pain points that will move these business KPIs. A sharp eye on business and customer outcomes will help you keep your focus and ensure that you are solving the right problems.

Rule 3: Harness the power of data

In the previous chapter, I spoke about the phenomenal opportunity for businesses to harness available data to understand customers deeply and offer them more personalized experiences. The key to successful optimization, whether of a customer journey or internal process, consists in analytically sound data-driven decisions. Research by Gartner has projected that by 2022, 90 per cent of corporate strategies will explicitly mention information as a critical enterprise asset and analytics as an essential competency.⁹ Effective use of data science is truly the 'secret sauce' in digital optimization. Algorithmic decision making based on data wins over human discretion. However, 90 per cent or more of the focus in data initiatives, both in terms of time and money, is usually on getting the data prepared and bringing it together in a data lake and less so on utilizing data for drawing out insights and for decision making. Reverse the approach: first, figure out the decision you want to make, then go about collecting data for it. As you anchor around the problem you are trying to solve (driven by your business KPIs), you will realize that your data management efforts will also become more focused and manageable.

Rule 4: Adopt an integrated, end-to-end approach

Large enterprises are shackled in siloed structures that are a function of their legacy businesses. This becomes a huge challenge when they embark on digital programs, as digital by its very nature is interdisciplinary. You need to put in place new structures that allow simplified cross-functional collaboration, swift decision-making and an integrated approach. Your operating model needs to enable 'aligned autonomy', ensuring that end-to-end capabilities are brought together and each team is empowered and vested with ownership of the outcomes. At the same time there needs to be clarity and alignment across teams on the business outcomes to be achieved.

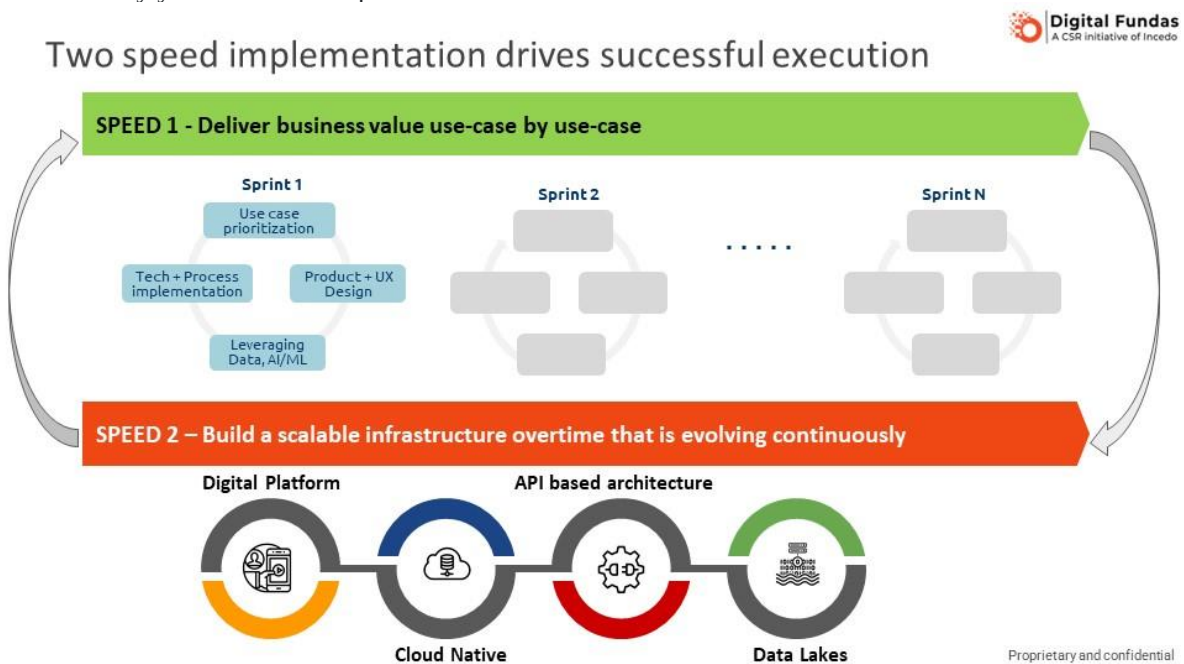
Many consumer internet companies have solved this difficulty by empowering business success managers or product managers who have end-to-end ownership of and accountability for initiatives. The product manager plays the role of an integrator who takes a customer-centric view, understands both business and technology, and works as a connector bringing together the different capabilities and initiatives required to achieve the desired outcomes.

Rule 5: Use a two-speed implementation approach

Going 'Big Bang' is, in my experience, a recipe for disaster, given the iterative nature of digital. At the same time, focusing on narrow, ad hoc use cases has limited impact. I strongly advocate the adoption of a 'two-speed' approach – focusing on high-impact use cases that will really move the needle to begin the journey while also leveraging the successes to build up your data and technology platform. Over successive agile implementation cycles, a two-speed strategy will allow you to solve for the most pressing business problems while developing a scalable platform.

Two-speed implementation is relevant not just for technology programs but also for business strategy.

Adopting multi-year, fixed business strategies is a kiss of death in a world where customer needs and available technologies are both changing very rapidly. In this paradigm, business strategy should be agile and driven by your execution experience.



Nitin's Conclusion:

Organizations will achieve success in their digital transformation journey only if they recognize that digital businesses are fundamentally different from traditional ones and need a different response. Digital business requires end-to-end business transformation, starting with a compelling vision followed by smart ways of ensuring successful execution. Merely approaching digital transformation as a technology project will yield only limited results.

Section 1

New Rules of Business

Chapter 5

Maximizing Returns from Digital Technology Investments: Part One - Why the Problem Persist

Key Takeaways

4 key reasons why Digital initiatives do not deliver results for businesses

Lesson 1: Why problems persist for companies embarking on digital transformation

Four Tech challenges why Digital initiatives do not deliver their promise  **Digital Fundas**
A CSR initiative of Incedo



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When it comes to digital transformation, most executives in large enterprises are simply not getting to the root of their problems. In this chapter, I will focus on the technology root causes of failed digital initiatives. In my experience of working with large legacy enterprises, I have seen four common root causes.

- Data infrastructure incapable of handling big data: The existing data infrastructure has limitations when it comes to leveraging unstructured data to drive insights. Originally created to deal with structured data, this infrastructure wasn't prepared for the explosion of big data. The pieces are hard to change once defined since the addition of new data dimensions requires expensive design and reprocessing effort. Moreover, legacy data infrastructure makes it very hard to get to a deeper level of granularity of insights. For example, when have you last seen a bank as precise at personalization as Amazon?
- IT processes do not support experimentation and agile methods: Traditional IT deployment cycles, as they are currently set up in many legacy enterprises, are not designed to support and drive the 'test, learn, adapt' paradigm of modern analytics. These processes should be based on experimentation. If something has to fail, it has to fail fast. You try something, measure the results, fail, change things, try again. Instead, in a legacy enterprise, you have to first spend a lot of time doing the analysis and making sure you get it right the first time, because if you fail it may be fatal.
- Back-end systems cannot handle real-time transactions: In most legacy enterprises, these systems end up being a bottleneck. They simply aren't designed to handle probabilistic choice-based workflows to enable real-time responsiveness to customer signals. Have you tried opening a bank account online? It's not possible, precisely because in most banks the back-end systems are not able to support this service or deliver real-time customer services online. As a result, businesses are unable to realize the true potential of tightly coupled systems and automation for real-time customer services and end-to-end transformation.
- Legacy business processes are too slow given changing expectations in the digital age: The basic 'mindset' of these processes is all about control, risk management and being comprehensive. These processes were designed to get things right the first time without fail. As a result, businesses have built very complex processes to avoid failure. But, as with anything very complex, it is hard for it to be fast in real-time. The legacy mindset of the 'right-the-first-time' process design makes it difficult to drive iterative experimentation-based improvements. In other words, even if the technology works, business processes end up slowing things down.

Nitin's Conclusion:

To achieve lasting change, I believe legacy companies need to stop playing defence and start playing offence. They should take lessons from the digital native companies that have leapfrogged them by reimagining these dimensions.

The main point is that digital initiatives themselves should be leading strategy and not the other way around. This will allow businesses to drive speed to market, unlock ROI and – most importantly – change the game from one of defence to offence.

Section 1

New Rules of Business

Chapter 6

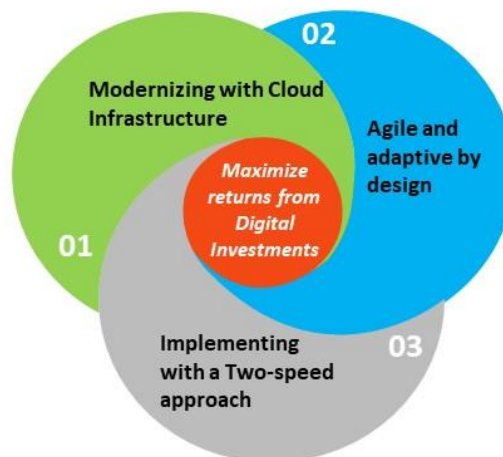
Maximizing Returns from Digital Technology Investments: Part Two: Play Offence, not Defence

Key Takeaways

Critical shifts to make to maximize returns on Digital Technology Investments

Lesson 1: Learnings for Legacy Enterprises

Changing the game in Technology Investments from Defense to Offense requires three critical shifts



Some specific and actionable best practices from digital natives that can help legacy enterprises significantly increase ROI from their digital technology investments:

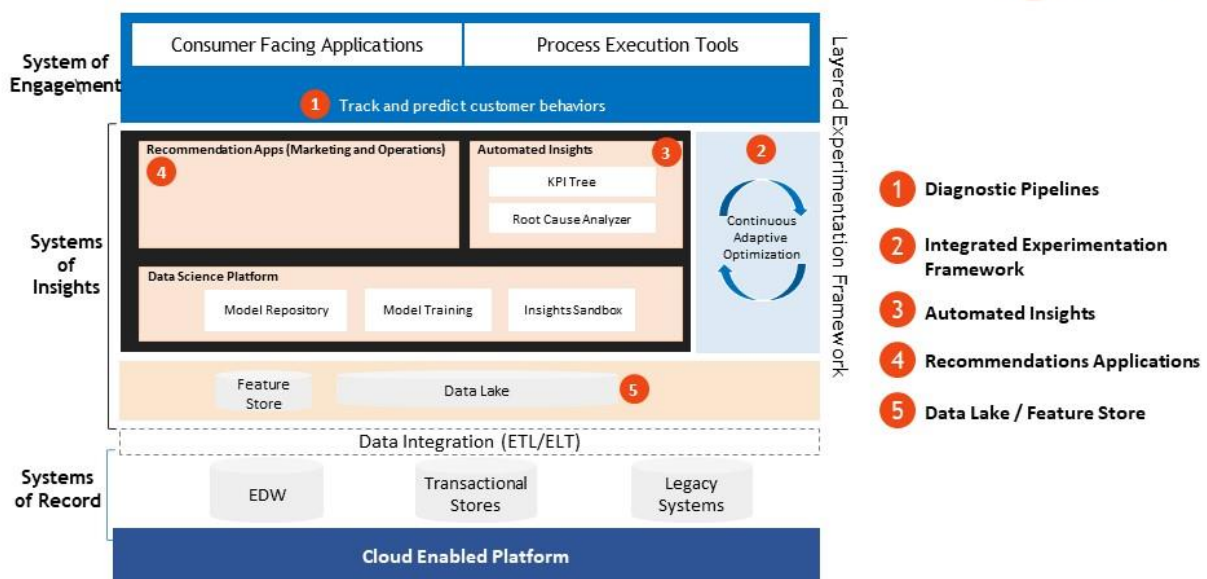
A. Cloud-native architecture

This architecture is an essential piece of the puzzle, and luckily for us, it is already here. In recent years, cloud strategy has moved from enabling compute resources to providing an integrated platform as a service (PaaS).

When fully realized, cloud standardizes technical skill sets, creating talent acquisition leverage and enabling improved development process, further accelerating time to market. Leveraging AWS, Azure and GCP (Google Cloud) provides the best-of-breed infrastructure, with the added advantage that these cloud providers continue to evolve and innovate at a rapid pace. That means enterprises can configure highly scalable end-to-end solutions quickly. So, what businesses need to do is to develop comprehensive cloud application and data migration strategies.

The best practices of a native cloud-based architecture would include the following five components:

Five extensions of a cloud - native architecture that drive speed-to-market and unlock ROI



i. Diagnostic pipelines

Business infrastructure must apply a comprehensive approach to track, monitor, predict and experiment with elements such as customer journeys from beginning to end, multi-channel funnel continuity, business process throughput, marketplace dynamics and data models.

This is important because engineering scalable, real-time data capture pipelines and integrating data capture into a data lake are critical for realizing business impact.

ii. Integrated experimentation framework

Business infrastructure and applications need to support experimentation to enable an innovation cycle. That means establishing clear business baselines, measuring KPI movements in a statistically significant manner, and integrating experimentation in the change management process to measure and validate impacts.

Operationalized data science cannot be enabled without a continuous experimentation cycle. So, it is important to both embed

experimentation into all layers of business infrastructure and develop an experiment-driven product management process.

iii. Automated insights

Traditional analytics approaches have created significant 'dashboard fatigue' in organizations because of their non-scalable manual tooling and methodologies. Ad hoc analysis requires significant effort and often leads to new development efforts. Aggregate views tend to lead to an extremely rigid approach to analysis, limiting analytics approaches and thinking. New out-of-the-box analytics technologies often fail to live up to the hype because they lack business-specific models.

To overcome these challenges, insights processes must be automated. That means applying a data science-driven analytics approach by leveraging KPI tree composition, insights tools and business-specific models. This will allow the organization to effectively automate key aspects of the insights process.

iv. Recommendation applications

Recommendation engines (REs) drive real-time, autonomous decision making for businesses. They have moved beyond customer personalization to enable automation across infrastructure. REs can factor in a broad set of features to drive business outcomes and customer experience. A comprehensive data strategy – driven by KPI capture, experimentation, scalable data pipelines and business-focused data science – enables effective model embedding.

v. Data lake/feature store

A harmonized, consolidated data strategy is a critical requirement for operationalizing data science and evolving from the loose, semantic agreement of relational systems.

This strategy must include a consolidated view of structured and unstructured data types, semantic schema to enable federated cross- organization integration, insights from the sandbox environment for developing data science, feature stores to drive cataloguing and reuse of semantically consistent data, and data governance strategy driven by automation integrated into operations.

B. Agile and adaptive by design

Modern infrastructures must be highly adaptive, agile and measured. Continuous adaptation and experimentation must be built into the strategy. The good news is that cloud architecture allows new development models for maintaining continuity while integrating change. Agile architectures compose and avoid system/team monoliths, while adaptive portfolio management process is critical to success.

Experimentation and measurement drive impact-based prioritization. In other words, an agile mindset must be embraced by the organization to yield benefits. Outcomes must be continually measured and evaluated in order to keep up with changing business conditions.

C. Two-speed approach

So, what is the best and most effective, practical way for legacy enterprises to tackle digital initiatives? Despite my recommendation that they play offence and not defence, going all-out or adopting a 'big bang approach' might not be the right strategy in most cases. This is because of the iterative nature of digital; you need to allow time for experimentation, learning and course correction. On the other hand, focusing on narrow use cases would amount to playing catch-up and thus would have little to no impact. So, it is critical to develop your 'offence' approach thoughtfully.

Based on my experience of working with legacy enterprises, I recommend that businesses adopt a two-speed approach.

Speed one would involve solving specific business and customer problems and delivering direct impact. The specific projects and problems to solve would be based on your specific business priorities. Importantly, these projects have to be executed with an agile implementation mindset. **Speed two** means building a scalable platform and long-term capability enhancement. Here, businesses have to ensure collaborative discovery and reuse across projects, cross-business-unit cataloguing and sharing of artifacts, as well as adoption of common standards for various processes ranging from problem definition to solution deployment.

Nitin's Conclusion:

Approaching digital transformation as a one-off technology project is a waste of time and resources; this is what legacy enterprises have to realize. It will not produce the impact desired. The only way legacy enterprises can succeed in their digital transformation journey is by following the lead of the digital-native companies, which have been built fundamentally differently. Digital transformation has to happen end to end. It should start with a vision, followed by a smart set of technology choices, and then careful and strategic implementation of those choices with the two-speed approach